

SHUNGNAC, AK, USA

WMO: 701719

Lat: 66.888N

Lon: 157.162W

Elev: 63

StdP: 100.57

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 00490

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.2	(c) -40.4	(d) -43.2	(e) 0.1	(f) -37.5	(g) -39.2	(h) 0.1	(i) -32.3	(j) 15.5	(k) -7.3	(l) 14.4	(m) -6.2	(n) 2.3	(o) 110	(p) 0.775

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	26.0	15.3	23.7	14.5	21.8	13.8	16.7	23.5	15.6	21.5	14.7	20.0	3.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.1	17.9	13.1	9.5	16.8	12.4	9.0	16.1	46.8	23.3	43.8	21.5	41.2	20.0	20.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.2	(b) 11.5	(c) 10.0	(e) -42.4	(f) 28.5	(g) 4.3	(h) 2.5	(i) -45.4	(j) 30.3	(k) -47.9	(l) 31.8	(m) -50.3	(n) 33.2	(o) -53.4	(p) 35.0		
(a) 13.2	(b) 11.5	(c) 10.0	(e) -37.7	(f) 17.7	(g) 4.2	(h) 1.1	(i) -40.7	(j) 18.5	(k) -43.2	(l) 19.1	(m) -45.6	(n) 19.7	(o) -48.7	(p) 20.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	-20.2	-16.2	-15.9	-5.3	5.3	13.7	15.5	12.2	6.5	-2.6	-15.0	-18.7		
	DBStd	15.08	12.86	9.72	7.75	8.10	5.95	3.76	3.37	3.25	4.15	5.92	8.68	10.07		
	HDD10.0	5270	936	733	803	461	163	12	2	15	117	390	751	889		
	HDD18.3	7922	1194	967	1061	710	403	144	101	191	354	648	1001	1148		
	CDD10.0	407	0	0	0	1	18	122	171	83	12	0	0	0		
	CDD18.3	18	0	0	0	0	0	5	12	1	0	0	0	0		
	CDH23.3	235	0	0	0	0	1	72	158	4	0	0	0	0		
	CDH26.7	51	0	0	0	0	0	10	41	0	0	0	0	0		
(14) Wind	WSAvg	3.8	4.5	4.7	3.6	3.9	3.5	3.1	2.7	3.0	3.4	4.4	4.1	4.8		
(15) Precipitation	PrecAvg	425	27	28	21	15	19	36	57	67	50	41	33	31		
	PrecMax	547	63	76	63	54	34	98	100	156	107	114	85	70		
	PrecMin	320	3	2	2	2	5	12	33	26	23	14	2	9		
	PrecStd	68	16	18	13	11	9	16	17	25	24	21	24	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	3.0	4.1	13.5	22.9	28.2	30.6	23.5	18.8	10.6	3.3	1.4		
		MCWB	2.3	-0.7	-0.4	6.2	12.4	15.9	17.7	15.5	11.6	4.4	0.5	-0.3		
	2%	DB	1.3	1.0	1.2	10.5	20.2	25.4	27.4	21.6	16.3	7.7	0.1	0.0		
		MCWB	-0.3	-0.9	-2.2	4.3	10.8	14.6	16.0	14.1	10.0	3.5	-1.4	-1.5		
	5%	DB	-1.4	-0.7	-1.3	7.8	17.2	23.4	24.7	19.7	14.3	6.4	-2.0	-2.2		
		MCWB	-3.0	-2.2	-3.4	2.9	9.6	13.6	15.4	13.5	9.2	3.4	-3.2	-3.4		
	10%	DB	-3.8	-2.5	-4.0	5.4	14.7	21.4	22.4	17.9	12.6	4.9	-3.9	-4.9		
		MCWB	-5.0	-4.1	-5.8	1.6	8.4	12.9	15.0	12.8	8.9	2.5	-5.1	-5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	0.7	0.1	6.6	13.3	16.8	18.4	16.5	12.5	6.2	0.6	0.3		
		MCDB	4.8	1.5	3.7	13.2	21.8	24.0	26.8	21.7	16.5	7.6	3.0	0.8		
	2%	WB	0.3	-0.8	-1.8	4.3	11.6	15.5	17.4	15.2	11.3	4.9	-1.1	-1.3		
		MCDB	1.8	0.9	0.8	9.8	18.9	22.9	24.9	19.3	14.4	6.7	0.0	0.1		
	5%	WB	-2.2	-2.2	-3.5	3.0	10.3	14.7	16.5	14.2	10.3	3.7	-3.0	-3.1		
		MCDB	-1.2	-0.5	-1.1	7.4	15.5	21.5	23.1	18.3	13.1	5.4	-2.0	-1.8		
	10%	WB	-4.4	-4.1	-5.6	1.8	9.1	13.8	15.4	13.4	9.4	2.7	-4.9	-5.6		
		MCDB	-3.0	-2.6	-4.0	5.2	14.0	19.9	20.9	16.8	12.2	4.6	-3.9	-4.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	8.0	10.5	9.7	10.6	10.4	8.8	7.8	7.6	6.0	6.2	6.9		
		MCDBR	4.9	6.3	10.7	9.8	14.7	13.6	14.0	10.4	10.5	5.6	5.5	5.6		
		MCWBR	4.1	5.2	8.5	6.0	7.4	5.4	5.6	5.0	5.7	3.5	4.8	5.2		
	5% WB	MCDBR	4.9	6.4	10.3	8.9	12.6	11.5	12.0	8.4	8.5	4.9	5.4	5.7		
		MCWBR	4.2	5.4	8.2	5.6	6.3	5.0	5.2	4.3	5.0	3.5	4.8	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.241	0.260	0.322	0.355	0.364	0.367	0.353	0.306	0.257	0.169	N/A		
	taud		1.913	2.127	2.238	2.132	2.253	2.362	2.378	2.437	2.514	2.282	1.939	N/A		
	Ebn at Noon		212	607	807	826	837	841	824	799	763	587	202	0		
	Edh at Noon		13	50	76	113	112	104	100	85	61	42	12	0		
(44) All-Sky Solar Radiation	RadAvg		0.08	0.66	2.46	4.68	5.45	5.70	4.58	3.42	2.08	0.78	0.16	0.00		
	RadStd		0.01	0.06	0.17	0.24	0.54	0.51	0.38	0.43	0.27	0.12	0.02	0.00		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page